

PHYS205 Experiment 1 Lab Report: Standing Waves on a String

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1 Objective

In this experiment, you will explore the relationship between string length, wavelength, frequency, linear density, and string tension in a standing wave, thus gaining an empirical understanding of the normal modes of vibration in a stretched string.

2 Theory

The properties that characterize a wave are its wavelength λ , its period T , its frequency of oscillation f (measured in hertz (Hz), or $\frac{1}{s} = s^{-1}$), and its speed v . These properties are related by the equation:

$$\lambda = v.T = \frac{v}{f} \tag{1}$$

Mechanical waves propagate through a medium in either longitudinal or transverse modes. In longitudinal waves, the particles oscillate in the same direction as the wave's propagation, as seen in a slinky spring or sound waves. In contrast, transverse waves have particles oscillating perpendicular to the wave's direction, as in a stretched string.

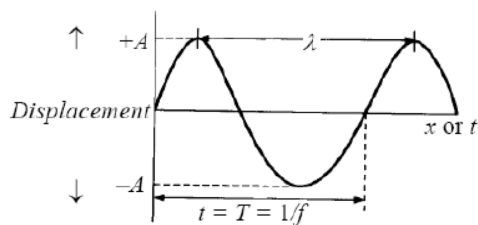


Figure 1: Wave Properties

As each particle oscillates, its maximum vertical displacement is the wave's amplitude, denoted as $+A$ or $-A$. Figure 1 shows displacement versus position or time. The wave's energy is directly proportional to its amplitude. The oscillation period is inversely proportional to the frequency, $T = 1/f$.

When two waves meet, they interfere, creating a simple superposition. If two waves moving in opposite directions have the same amplitude and frequency, their interference forms a standing wave, as shown in Figure 2. Destructive interference occurs at *nodes*, where displacement is minimum, while constructive interference occurs at *antinodes*, where displacement is maximum. The length of one segment of the standing wave is half its wavelength.

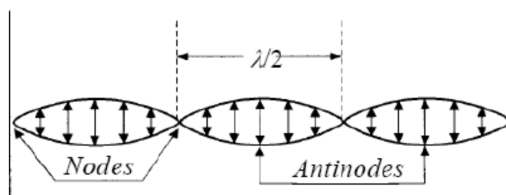


Figure 2: Standing Wave

When a string vibrates at one end, waves from the vibrator interfere with waves reflected from the opposite fixed end, creating standing waves at specific frequencies depending on the string's density, tension, and length. Vibrating the string at multiples of this frequency produces standing waves with multiple segments, known as *harmonics*.

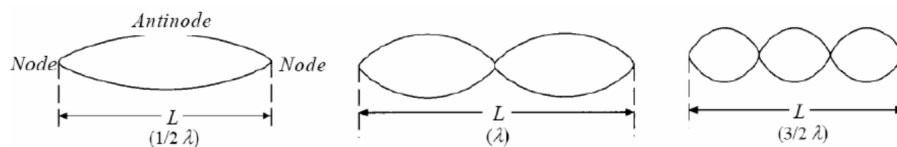


Figure 3: Wave Harmonics

For a wave on a string, the speed of the wave is also influenced by the tension in the string and its linear density. The relationship can be expressed as:

$$v = \sqrt{\frac{F_T}{\mu}} \quad (2)$$

Here, F_T represents the tension force, which is the elastic property of the string, while μ is the linear density, which is the mass per unit length of the string.

The tension applied by the hanging mass equals its weight (mg). In this experiment, the frequency is adjusted so the wave vibrates in four segments, ensuring the string length equals two wavelengths ($L = 2\lambda$).

In this case $F = mg$, and $L = 2\lambda$; these equations can be combined with equations (1) and (2):

$$f^2 = \frac{4g}{L^2\mu}m \quad (3)$$

3 Procedure

Equipment

- PC and Universal Interface
- 1 Mechanical Wave Driver
- 1 Electronic Balance
- 1 Roll of White String
- 1 Mass Set with Hanger
- 2 Banana Plug Patch Cords

- 1 Table Clamp and Pulley System
- 1 Support Rod
- 1 Meter Stick

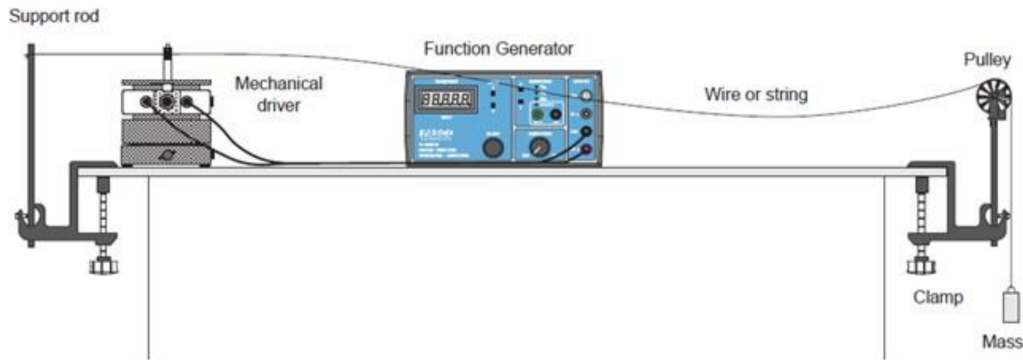


Figure 4: Experimental Setup

Part A: Inelastic String

A mass was hung from a string over a pulley created tension. A mechanical wave generator produced transverse waves in the string. Adjusting tension and frequency formed standing waves, as shown in Figure 4.

The experimental procedure was as follows:

1. Cut a piece of string approximately 2.5 m long and measure its length, L . Measure the mass of the string, m , using an electronic balance. Record both values in Table ??.
2. Calculate the string's linear mass density, $\mu = \frac{m}{L}$, and record it in Table ??.
3. Cut a 2 m portion of the string and tie a loop at each end. Attach one loop to a vertical support rod clamped to the mechanical wave driver, and pass the other end over a pulley approximately 1.5 m away. Attach a mass hanger to the free loop.
4. Use banana-plug patch cords to connect the driver to the signal generator output.

5. Hang a 50 g mass from the string over the pulley and record the total hanging mass in Table 2.
6. Measure the vibrating length of the string between the wave driver and the top of the pulley. Record this value ($L = 1.5$ m) in Table 2.
7. Turn on the signal generator and adjust the sine wave frequency f until four antinodes are observed. Record the frequency in Table 2.
8. Increase the hanging mass in 50 g increments up to 250 g, repeating Step 7 for each case and recording the corresponding frequencies.

Part B: Elastic String

1. Repeat the same procedure described in Part A, this time using an elastic string. Record all measurements and calculated values in the corresponding tables (Table 3 or Table 4).
2. Measure the extension of the elastic string when the maximum mass (250 g) is suspended. Record this value below.

4 Data

Part A: Inelastic String

Table 1

Quantity	Value
Length of String (L)	2.20 m
Mass of String (m)	7.8×10^{-4} kg
Linear Density (μ)	3.5×10^{-4} kg.m ⁻¹

Table 2

f (Hz)	m (kg)
52	0.05
74	0.1
90	0.15
104	0.2
116	0.25

Length of String (L) = 1.5 m

Part B: Elastic String

Table 3

Quantity	Value
Length of String (L)	1.85 m
Mass of String (m)	5.76×10^{-3} kg
Linear Density (μ)	3.1×10^{-3} kg.m ⁻¹

Table 4

f (Hz)	m (kg)
20	0.05
30	0.1
38	0.15
46	0.2
51	0.25

Length of String (L) = 1.5 m

Stretched length at maximum mass ($L_{stretched}$) = 2.7 m

5 Data Analysis

In these analyses, the acceleration due to gravity was assumed to be $g \approx 9.81$ m/s², and the string length was measured as $L = 1.5$ m. These values were used in the calculations.

Part A: Inelastic String

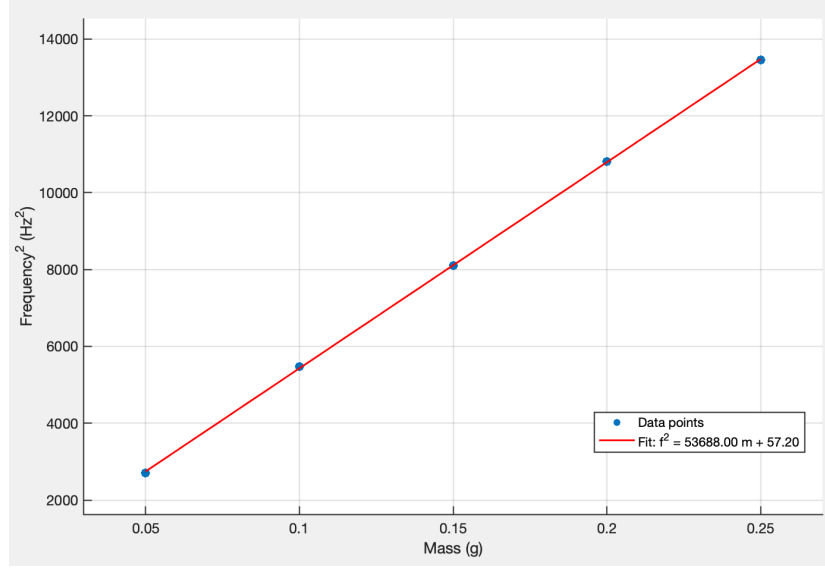


Figure 5: $f^2 - m$ graph

The best-fit line was obtained using MATLAB, and its slope was found to be $53,688 \text{ Hz}^2 \cdot \text{kg}^{-1}$. Then, Equation (3) can be rewritten as

$$\mu = \frac{4g}{L^2} \frac{m}{f^2} = \frac{4g}{L^2} \frac{1}{\text{Slope}} \quad (4)$$

By substituting the measured values into Equation (4), the experimental result for the linear density is found as $\mu = 3.25 \times 10^{-4} \text{ kg/m}$. Now, the actual (measured) value given in Table 1 can be compared with the experimental result:

$$\mu_{exp} = 3.25 \times 10^{-4} \text{ kg/m}$$

$$\mu_{actual} = 3.5 \times 10^{-4} \text{ kg/m}$$

$$\text{The Percent Error} = \frac{|\mu_{actual} - \mu_{exp}|}{\mu_{actual}} \times 100 = 7.14\%$$

In addition, the speed of the standing wave in minimum (0.05 kg) and maximum (0.25 kg) hanging masses can be calculated using the linear density of

the string and Equation (2). F_T in Equation (2) is equal to mg so it can be rewritten as:

$$v = \sqrt{\frac{mg}{\mu}} \quad (5)$$

When these variables are substituted, $v_{min,calc} = 37.44 \text{ m/s}$ and $v_{max,calc} = 83.71 \text{ m/s}$ using μ_{actual} .

These speed values can also be calculated using Equation (1) and values in Table 2. The Equation (1) is rearranged as $v = \lambda f$. Since the frequency was adjusted such that the wave vibrates in four segments in this experiment, the length of the string is always equal two wavelengths ($L = 2\lambda$), so $\lambda = 0.75 \text{ m}$. Now, when the variables are substituted with $\lambda = 0.75 \text{ m}$ and data in Table 2, $v_{min,exp} = 39 \text{ m/s}$ and $v_{max,exp} = 87 \text{ m/s}$. Now, the percent error can be calculated:

$$v_{min,calc} = 37.44 \text{ m/s} \quad v_{min,exp} = 39 \text{ m/s}$$

$$\text{The Percent Error}_{min} = \frac{|v_{min,calc} - v_{min,exp}|}{v_{min,calc}} \times 100 = 4.17\%$$

$$v_{max,calc} = 83.71 \text{ m/s} \quad v_{max,exp} = 87$$

$$\text{The Percent Error}_{max} = \frac{|v_{max,calc} - v_{max,exp}|}{v_{max,calc}} \times 100 = 3.93\%$$

Part B: Elastic String

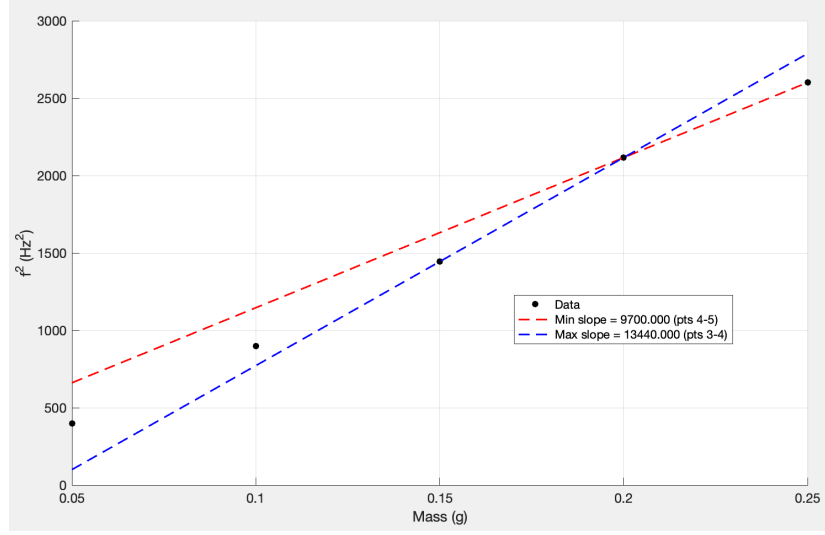


Figure 6: $f^2 - m$ graph

Even though this graph looks like linear, it cannot be linear because an elastic string was used in this part of the experiment, and its linear density changes depending on the hanging mass. The max and min slopes was found using MATLAB with calculating the slope between each two pair using iteration, and the min slope was found to be $9,700 \text{ Hz}^2.\text{kg}^{-1}$ between points 4 and 5, and the max slope was found to be $13,440 \text{ Hz}^2.\text{kg}^{-1}$ between points 3 and 4.

To estimate the stretched density ($\mu_{stretched}$) of the cord, the the unstretched density ($\mu_{unstretched}$) and length ($L_{unstretched}$), μ and L in Table 3 respectively, can be used.

$$\mu_{stretched} = \mu_{unstretched} \cdot \frac{L_{unstretched}}{L_{stretched}}$$

When this equation is calculated to find $\mu_{stretched}$ with values given in tables above, $\mu_{stretched} = 2.12 \times 10^{-3}$

The density can also be calculated using the min and max slopes with the Equation (4). The density with min slope ($\mu_{minSlope}$) is equal to 1.8×10^{-3} and the density with max slope ($\mu_{maxSlope}$) is equal to 1.3×10^{-3} after calculation. Then, this result can be easily seen:

$$\mu_{stretched} > \mu_{minSlope} > \mu_{maxSlope}$$

but the expectation is the density decreases while stretching because the length increasing and $\mu \propto \frac{1}{L}$. Some measurement mistakes and non-ideal elastic string can cause this error.

Additionally, the speed of the standing wave in minimum (0.05 kg) and maximum (0.25 kg) hanging masses can be calculated using the densities by Equation (5).

$$v_{min,calc} = \sqrt{\frac{0.05g}{\mu_{unstretched}}} = 12.58 \text{ m/s} \quad v_{max,calc} = \sqrt{\frac{0.25g}{\mu_{stretched}}} = 34.01 \text{ m/s}$$

To calculate the experimental value of the speeds, the $v = \lambda f$ formula can be used with the same λ value as in Part A. In this formula, λ can be substituted with 0.75 and f can be substituted with the corresponding values in the Table 4.

$$v_{min,exp} = 15 \text{ m/s} \quad v_{max,exp} = 38.25 \text{ m/s}$$

Now, the percentage error can be calculated to see the accuracy of the experiment.

$$\text{The Percent Error}_{min} = \frac{|v_{min,calc} - v_{min,exp}|}{v_{min,calc}} \times 100 = 19.24\%$$

$$\text{The Percent Error}_{max} = \frac{|v_{max,calc} - v_{max,exp}|}{v_{max,calc}} \times 100 = 12.47\%$$

6 Conclusion

In this experiment, the relationship between the frequency, tension, and linear density of a oscillating string was studied with the standing waves. For the inelastic string, the $f^2 - m$ relationship was found highly linear, confirming the $f^2 = \frac{4g}{L^2\mu}m$ formula. The error between theoretical and the experimental densities are 7.14% This small error can be caused by the measurement uncertainties in string length and frequency readings.

In the elastic string experiment, the $f^2 - m$ graph is not linear. This behavior is expected, since the elastic string stretches under hanging masses, causing its linear density to change. The calculated densities from the minimum and maximum slopes and the expectation is the stretched density should be between these values, but it is not. The small inconsistency observed here

and it may result from non-ideal elastic behavior of string or experimental mistakes.

Finally, the comparison between the calculated and experimental wave speeds showed good consistency for both strings, with percentage errors about 4% for the inelastic string and below 20% for the elastic string. The error is bigger for elastic string because there are more factors that can affect the results in elastic string than inelastic string and dealing with the more factors causes higher errors.

Overall, the experiment successfully demonstrated how standing wave frequencies depend on both the tension and the linear density of a string, and how deviations from linearity can be explained by the elastic strings.